LLM USAGE ANALYTICS PLATFORM
TELEMETRY INSIGHTS FROM DEVELOPER ACTIVITY

Understanding LLM Usage Through Telemetry Data

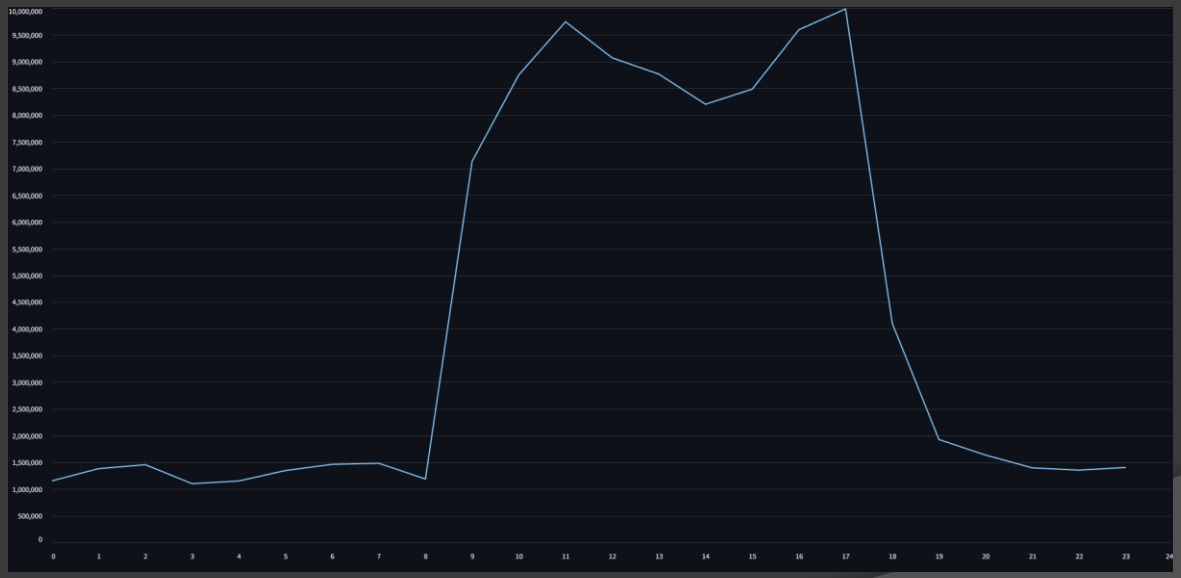
- Claude Code generates telemetry logs for developer interactions
- Logs include information about:
  - API requests
  - token usage
  - model selection
  - tool execution
- **Goal:** transform raw telemetry logs into actionable insights about usage patterns and costs.

Analytics Pipeline

Analytics Platform Components:
Telemetry ingestion pipeline • Structured dataset generation
SQL analytics layer • Interactive Streamlit dashboard

Developer Usage Patterns

- LLM usage follows clear **working-hour patterns**
- Most activity occurs between **10:00–17:00**
- Token usage varies significantly across developers
- Overall usage in the dataset generated **~$6000 in LLM cost**

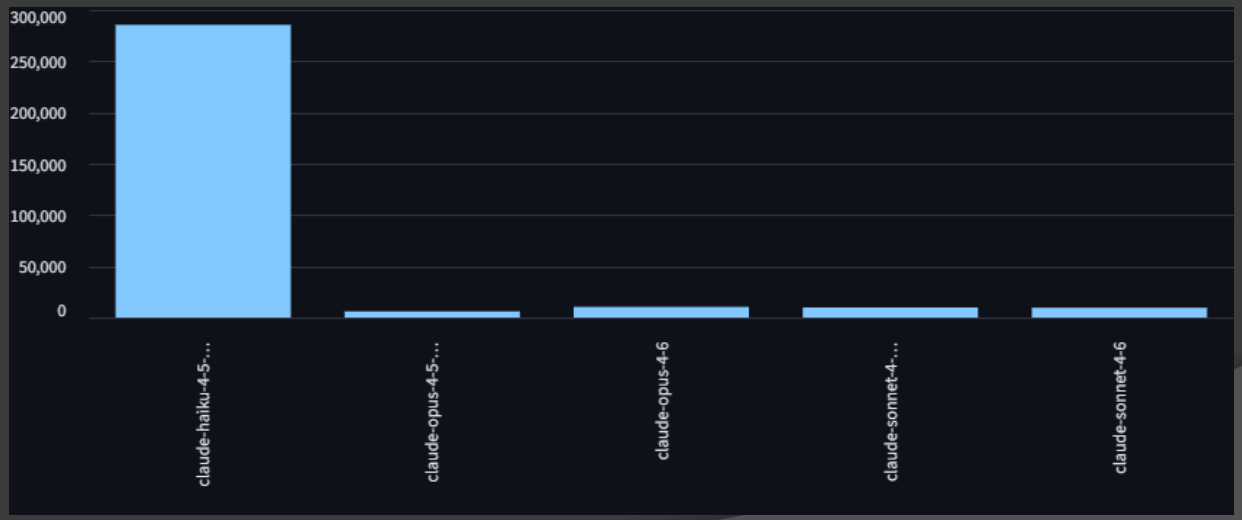


Hourly Token Usage

Model Efficiency Analysis

- **Models differ significantly in tokens-per-dollar efficiency**
- Choosing the right model can significantly reduce cost
- Efficiency Metric:

  Tokens per Dollar = Total Tokens / Total Cost
- Different models show noticeable efficiency differences, which can impact cost optimization strategies.



Model Efficiency (Tokens per Dollar)

Tool Usage in Developer Workflows

- Tools are frequently used alongside LLM requests
- Most tool executions complete successfully
- Tools like **Read** and **Bash** dominate usage

Tool	Calls	Success Rate
Read	45,077	98.51%
Bash	42,332	93.04%
Edit	18,759	99.11%
Grep	11,342	98.96%
Glob	6,929	99.03%

Top 5 Most Used Tools

- Tool integrations appear to be a core part of developer workflows